Strong unextendibility of small Pauli MUB sets: exact bounds for two, three and four qubits

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Abstract

A set of mutually unbiased stabiliser bases (Pauli MUBs) is *weakly unextendible* if no further Pauli class is unbiased to all of them, and *strongly unextendible* if no vector at all is. For a set $\mathcal{S}$ in dimension d we study $\mathcal{F}(v) = \sum_e \big(|\langle e|v\rangle|^2 - |v|^2/d\big)^2$, the sum over all basis vectors e of the set; strong unextendibility is $\min_{|v|=1} \mathcal{F} > 0$. We prove exact, quantitative lower bounds. For two qubits, the minimum is exactly $1/4$. For three qubits, all 24 192 five-element sets form one Clifford orbit and $\min \mathcal{F} = 1/8$ exactly, attained by stabiliser states; the proof conditions on a distinguished uncovered Pauli and closes a $4 \otimes 4$ inequality, which resists low levels of the SOS/PPT hierarchy, with a two-moment eigenvalue bound. For four qubits, the smallest weakly unextendible sets have exactly five elements (an exhaustive search, matching Grassl's size $q^{n/2}+1$). There are 105 283 584 of them in two types. One type satisfies $\mathcal{F} \geq |v|^4/80$ exactly, so five Pauli-class bases are strongly unextendible in $d = 16$ (general, non-Pauli MUB triples are already strongly unextendible in many dimensions [5]), and its minimum (≈ 0.0524) lies below its stabiliser value $1/16$. For five qubits the smallest such sets have exactly 9 elements (an exhaustive computer search excludes sizes up to 8, and one of size 9 exists). The proofs combine a twirl identity, the graph-theoretic uncertainty relation of de Gois, Hansenne and Gühne, Pauli-symmetry reduction of sum-of-squares certificates, facial reduction, and verification in exact rational arithmetic.

1 Setting

Let n qubits carry the Pauli group with symplectic space $W(2n-1, 2)$. Its points are Pauli operators up to phase, and its Lagrangians are maximal commuting classes. Disjoint Lagrangians give mutually unbiased stabiliser bases, and a weakly unextendible set is a maximal partial spread [1, 2]. Mandayam, Bandyopadhyay, Grassl and Wootters [1] constructed such sets with $d/2+1$ elements for $d = 4, 8$. They proved strong unextendibility for specific examples with a Gröbner basis computation, and noted that a general proof was open. Kantor [3] surveys maximal symplectic partial spreads and reports Grassl's sets of size $q^{n/2}+1$.

Twirl identity. For a stabiliser basis B_L, $\sum_{e \in B_L} |e\rangle\langle e|^{\otimes 2} = d^{-1} \sum_{P \in L} P^{\otimes 2}$. Hence, with C the covered Paulis and U the uncovered ones,

$$\mathcal{F}(v) = \frac{1}{d} \sum_{P \in C} \langle P \rangle^2 = \frac{1}{d}\Big(d - 1 - \sum_{P \in U} \langle P \rangle^2\Big) \qquad (|v| = 1), \tag{1}$$

since $\sum_{P \neq I} \langle P \rangle^2 = d - 1$ for pure states. A stabiliser state of a Lagrangian meeting C in m points gives $\mathcal{F} = m/d$. Maximality forces $m \geq 1$, so $\min \mathcal{F} \leq m^*/d$, where m^* is the least such m.

Lovász bound. De Gois, Hansenne and Gühne [4] prove $\sum_i \langle A_i\rangle^2 \le \vartheta(G)$ for dichotomic observables, where G is their anticommutation graph and ϑ the Lovász number [7]. Applied to U, $\min \mathcal{F} \ge (d-1-\vartheta)/d$. The bound is tight when $\vartheta = \alpha(G) = d-1-m^*$.

2 Results

Theorem 1 (two qubits)**.** *The three-element maximal partial spreads of $W(3,2)$ form one orbit, since exactly one passes through a fixed complementary pair (Witt), and for them $\min \mathcal{F} = 1/4$ exactly. Here $\vartheta = \alpha = 2$, and equivalently the six uncovered Paulis split into two pairwise-anticommuting triples.*

Theorem 2 (three qubits)**.** *The maximal partial spreads of $W(5,2)$ have sizes 5 (24 192 sets, one $\mathrm{Sp}(6,2)$-orbit) and 9 (complete sets). For every five-element set and every $v \in \mathbb{C}^8$, $\mathcal{F}(v) \ge |v|^4/8$, and stabiliser states attain $1/8$; so $\min \mathcal{F} = 1/8$ exactly. (An independent degree-6 SOS certificate gives $\mathcal{F} \ge (1/8 - 10^{-6})|v|^4$.) A weaker but structural bound is $\mathcal{F} \ge \frac{3-\sqrt{5}}{8}|v|^4$, from $\vartheta \le 4+\sqrt{5}$, certified over $\mathbb{Q}(\sqrt{5})$ in the rank-21 coherent closure of the uncovered graph.*

Proof of $\mathcal{F} \ge |v|^4/8$. By the twirl identity $\mathcal{F} = (7-\Sigma_U)/8$ for unit v, where $\Sigma_U = \sum_{P\in U}\langle P\rangle^2$ over the 27 uncovered non-identity Paulis; we show $\Sigma_U \le 6$. By the single orbit it suffices to treat one set. Let $P_0 \in U$ be the uncovered Pauli fixed by the stabiliser of the set, $V_\pm$ its eigenspaces, $v = c\psi_+ + s\psi_-$ with $\psi_\pm \in V_\pm$ unit and $t = \langle P_0\rangle = c^2 - s^2$.

Commuting part. The 15 uncovered Paulis B commuting with P_0 (other than P_0) map bijectively onto the points of $p_0^\perp/\langle p_0\rangle \cong W(3,2)$ (checked). On each $V_\pm$ they act, up to sign, as the 15 non-identity two-qubit Paulis, so two-qubit purity gives $\sum_B \langle\psi_\pm|P|\psi_\pm\rangle^2 \le 3$; since $\langle P\rangle = c^2\langle\psi_+|P|\psi_+\rangle + s^2\langle\psi_-|P|\psi_-\rangle$, convexity gives $\sum_B \langle P\rangle^2 \le 3$.

Anticommuting part. The 12 uncovered Paulis anticommuting with P_0 form six pairs $\{a_j, \pm i a_j P_0\}$ (checked). For such Q, $\langle Q\rangle = 2cs\,\mathrm{Re}\langle\psi_+|Q|\psi_-\rangle$, and a pair contributes exactly $(1-t^2)|\langle\psi_+|a_j|\psi_-\rangle|^2$. Writing $M_j : V_- \to V_+$ for the restriction of a_j (a unitary 4×4 matrix), $\sum_A \langle Q\rangle^2 = (1-t^2)\sum_{j=1}^6 |\langle\psi_+|M_j\psi_-\rangle|^2 \le (1-t^2)\,\kappa$, with $\kappa = \max_{|x|=|y|=1}\sum_j |\langle x|M_j y\rangle|^2$.

$\kappa \le 3$. Fix a unit $y \in V_-$ and let $W = [M_1 y, \dots, M_6 y]$, a 4×6 matrix. Then $\max_x \sum_j |\langle x|M_j y\rangle|^2 = \lambda_{\max}(WW^*)$. The 4×4 matrix WW^* is positive semidefinite with $\mathrm{Tr}\,WW^* = \sum_j |M_j y|^2 = 6$, and $\mathrm{Tr}(WW^*)^2 = \mathrm{Tr}(W^*W)^2 = \sum_{j,k} |\langle y|M_j^* M_k|y\rangle|^2$. The 15 products $M_j^* M_k$ $(j<k)$ are the restrictions to V_- of $a_j a_k$, whose classes $a_j + a_k$ are distinct modulo p_0 and different from $0, p_0$; so up to phase they are the 15 non-identity two-qubit Paulis on V_- (verified exactly: with the identity they are pairwise Hilbert–Schmidt orthogonal). Purity of y gives $\sum_{j<k} |\langle y|M_j^* M_k|y\rangle|^2 = 4-1 = 3$, hence $\mathrm{Tr}(WW^*)^2 = 6 + 2\cdot 3 = 12$ for every y. If $\lambda_1 \ge \cdots \ge \lambda_4 \ge 0$ are the eigenvalues of WW^*, then Cauchy–Schwarz on $\lambda_2, \lambda_3, \lambda_4$ gives $(6-\lambda_1)^2 \le 3(12-\lambda_1^2)$, that is $4\lambda_1(\lambda_1 - 3) \le 0$, so $\lambda_1 \le 3$ (the two-moment bound of Wolkowicz and Styan [6]). Equality forces the spectrum $(3,1,1,1)$, which the numerical maximiser has.

Therefore $\Sigma_U \le t^2 + 3(1-t^2) + 3 = 6 - 2t^2 \le 6$. □

Remark. The moment step uses only that the six operators are unitary and that their 15 pairwise quotients, with the identity, form part of an orthogonal unitary basis. Combinatorially, the classes of $a_j + a_1$ $(j = 1, \dots, 6)$ modulo p_0 form a six-subset of $\mathbb{F}_2^4$ containing 0 whose 15 pairwise sums are all the non-zero vectors. The five non-zero classes anticommute along a pentagon, and that alone forces the property: the pentagon's symplectic Gram matrix has kernel spanned by the all-ones vector, so the only linear relation among them is their sum. The inequality $\kappa \le 3$ is tight and has no decomposable (PSD plus partial-transpose) certificate, and the SOS/PPT hierarchy fails at

levels $(1,2)$, $(2,1)$ and $(2,2)$; the moment argument bypasses this. It uses only the property that the 15 pairwise sums of six labels are all the non-zero vectors of $\mathbb{F}_2^4$. There are 448 such six-element subsets of two-qubit Pauli labels, in three affine-symplectic orbits of sizes 16, 192 and 240. The bound $\kappa \le 3$ therefore holds on all of them, and it is attained on the orbits of sizes 192 (which contains the configuration above and has a PPT Choi matrix) and 240. On the orbit of size 16 (a twisted Werner–Holevo channel) $\kappa = 2$. The equality locus contains non-stabiliser states, e.g. $y_t \propto (1, it, -it, 1)$ for all real t.

Independent SOS method. $\mathcal{F}(v)|v|^2 - \delta|v|^6$ is written as an SOS in the degree-3 monomials and in the monomials $v_i v_j \bar{v}_k$. The Pauli group acts on both monomial spaces with centre -1, so each is a multiple of the unique 8-dimensional irrep, and invariant Gram matrices reduce to blocks of size 15 and 36. All coefficients are Gaussian rationals. At $\delta = 1/8 - \varepsilon$, a strictly interior certificate is rounded (denominator 16). Its pivot variables are restored exactly, every coefficient equation is verified in integer arithmetic, and positivity is checked by exact LDL^{T}, for $\varepsilon = 10^{-3}, \ldots, 10^{-6}$. At $\delta = 1/8$ the 160 stabiliser minimisers force the 15-block to vanish, and facial reduction leaves a 16-dimensional face with zero interior margin. This suggests that the degree-6 bound approaches $1/8$ without attaining it.

Theorem 3 (four qubits)**.** *(i)* $W(7,2)$ *has no maximal partial spread of size 3 or 4. The smallest size is 5, and there are* 105 283 584 *such sets. (ii) They fall into two invariant types. For a set of the first type (lines listed in the repository),* $\vartheta \le 74/5$*, so* $\mathcal{F} \ge |v|^4/80$ *exactly, while numerically* $\min \mathcal{F} = 0.0523626 < 1/16$*. A representative of the second type is a* lift *of a three-qubit five-set: for it, 120 uncovered points* p *project the set onto a maximal five-set of* $p^\perp/\langle p\rangle \cong W(5,2)$*, whereas the representative of the first type has no such point. It inherits a mirror partner (a second partition of the same covered set) and have numerical minimum* $1/16$*. (iii) For several inequivalent nine-element sets,* $\min \mathcal{F} = m^*/16$ *exactly (*$1/16$ *or* $1/8$*), via a facially reduced Lovász dual.*

Proposition 4 (five qubits)**.** $W(9,2)$ *has no maximal partial spread of size at most 7, and it has one of size 9 (verified independently). An exhaustive search over all eight* $O(5)$*-orbits of the second matrix (exact-cover branching with a valid covering prune; 433 517 130 nodes; controls reproduce the* $n = 3$ *sizes and the size-7 exclusion) finds no maximal partial spread of size 8. Hence the smallest five-qubit maximal Pauli MUB set has exactly 9 elements. (A second complete run with a different traversal order, on the two representatives that remain after the* S_3 *symmetry* $S \mapsto S^{-1}$*,* $S \mapsto S + I$ *of the three fixed Lagrangians, also finds none: 107 943 913 nodes. Both runs use the same code base.) The data* $3, 5, 5, 9$ *for* $n = 2, \ldots, 5$ *fit* $2^{\lceil n/2 \rceil} + 1$*. This is the size* $q^k + 1$ *of Kantor's maximal partial spreads in dimensions* $4k$ *and* $4k - 2$ *for* $q = 2$ *[3], and for even* n *it is the value in Grassl's conjecture on the smallest size, which Kantor reports to fail in dimension 8 for* $q = 2^{2e+1} > 2$*. The present results show that for* $q = 2$ *this value is the exact minimum in dimensions 6, 8 and 10; in particular Grassl's conjecture holds for* $\mathrm{Sp}(8,2)$*. Minimality for larger* n *is a conjecture here. The size-9 set has numerical minimum* $\mathcal{F} = 1/32$*, its stabiliser value. For* $n = 6$ *a coverage-biased greedy search finds a maximal partial spread of size* $9 = 2^3 + 1$ *(verified over all 888 832 nonsingular symmetric* 6×6 *matrices) and none smaller, consistent with the same pattern; this is evidence, not a proof. For the five-qubit size-9 set the Lovász number of the uncovered graph (744 vertices, 368-regular) satisfies* $30.340 \le \vartheta \le 30.449$ *(floating-point certificates: an eigenvalue-form dual and* $\vartheta(G)\vartheta(\bar{G}) \ge 744$*). Hence* $\min \mathcal{F} \ge 0.0172$*, and the Lovász route cannot reach* $1/32$*; as at* $d = 8$*,* $\vartheta > \alpha$*. Unlike type B at* $d = 16$*, the set is not a lift: no uncovered point projects it onto a partial spread of* $p^\perp/\langle p\rangle$*.*

Proposition 5 (regulus sets)**.** *Let* $q = 2^k$ *and regard* $2k$ *qubits as two* $\mathbb{F}_q$*-qudits, with symplectic form* $\mathrm{Tr}(x \cdot z' + z \cdot x')$ *on* $\mathbb{F}_q^4 \ni (x_1, x_2, z_1, z_2)$*. The* $q + 1$ *Lagrangians* $L_\infty = \{x = 0\}$ *and*

$L_t = \{z_1 = tx_2,\ z_2 = tx_1\}$, $t \in \mathbb{F}_q$, form a regulus of the hyperbolic quadric $Q = x_1z_1 + x_2z_2$; their eigenbases are the computational basis and $\mathrm{CZ}_t(F \otimes F)|x_1, x_2\rangle$ with $\mathrm{CZ}_t = \mathrm{diag}\,(-1)^{\mathrm{Tr}(tx_1x_2)}$. This is the explicit form of Kantor's sets of size $q^k + 1$ [3]. For $q = 2, 4, 8$ the set is maximal (checked against all nonsingular symmetric matrices; 888 832 for $q = 8$) and its covered Paulis are exactly the points with $Q = 0$. (i) With $A(x_1, x_2)$ the amplitudes and B the row-wise Fourier transform of A, $\mathcal{F} + (q+1)/q^2 = \sum_\ell \|\mathfrak{F}_\ell(B|_\ell)\|_4^4$, the sum running over the $q(q+1)$ affine lines of $AG(2, q)$ and $\mathfrak{F}_\ell$ being the additive Fourier transform on the line. (ii) For $q = 4$ the set is the lifted type of Theorem 3 (same invariants), and $\mathcal{F} + \frac{5}{16} = \sum_p |A_p|^4 + \frac{1}{4}|\sum_p A_p^2|^2 + \sum_\ell \sum_{ab|cd} |A_aA_b + A_cA_d|^2$, the last sum over the 20 lines of $AG(2, 4)$ and their three pairings. Numerically $\min \mathcal{F} = 1/16$; a Pauli-reduced degree-6 SOS gives $\mathcal{F} \geq 0.0452$ (floating point), and the PSD-plus-partial-transpose level gives only $\mathcal{F} \geq 0$. (iii) For $q = 8$, numerically $\min \mathcal{F} = 2.498509/64$ (300 starts, analytic gradients), which is not a stabiliser value. The quotients of this set at points with $\mathrm{Tr}\, Q = 1$ are maximal five-qubit sets of size 9, of at least two types, with numerical minimum $2.498509/32$. So minimal sets of the same size differ in strength: the five- and six-qubit sets found by search have $\min \mathcal{F} = 1/d$.

Method. Fixing the X- and Z-type Lagrangians, a Lagrangian disjoint from both is the graph of a nonsingular symmetric matrix S over $\mathbb{F}_2$, and two graphs are disjoint iff $S + S'$ is nonsingular. A maximal partial spread is thus a set of nonsingular symmetric matrices with nonsingular pairwise differences that leaves no nonsingular T with all $T + S_i$ nonsingular. For $n = 5$ there are 13 888 such matrices; $S_1 = I$ without loss of generality, and S_2 is reduced modulo the orthogonal group of order 720, followed by an exhaustive depth-first search with a valid covering prune for size 7. The same code reproduces the counts 56 ($n = 3$) and 896 ($n = 4$) found by direct enumeration.
Method for (i). By Witt's theorem, $\mathrm{Sp}(8, 2)$ is transitive on ordered pairs of complementary Lagrangians, so every candidate can be taken to contain a fixed pair. The remaining one or two Lagrangians are enumerated exhaustively. As a control, the same code finds no sizes 3 and 4 for three qubits, and finds 56 five-sets through a fixed pair, matching $24\,192 \cdot 20/(135 \cdot 64) = 56$. *Method for (iii).* The maximum independent sets of the uncovered graph are the Lagrangians meeting C in m^* points, minus those points. Complementary slackness with them gives linear equations on the dual. Imposing these equations makes rounding succeed, and the certificate is checked by exact LDL^{T}.

Remark (two qutrits). For the 432 five-element maximal partial spreads of $W(3, 3)$, $\mathcal{F} \geq |v|^4/9$ exactly, via an S_6/Jucys–Murphy identity. The true minimum is numerically $(207 - 19\sqrt{57})/432 \approx 0.14711$, which is an identification, not a proof. Mirror partners there lie in opposite orbits and are exchanged only by antiunitaries; a third-moment witness of size 16/27 separates them. In characteristic 2 the corresponding partners are Clifford-equivalent.

3 Summary and open questions

d	smallest size	$\min \mathcal{F}$	status
4	3	$1/4$	exact
8	5	$1/8$	exact
9	5	$\geq 1/9$; ≈ 0.14711	exact bound; numerical value
16	5	$\geq 1/80$; ≈ 0.05236 (one type)	exact bound; numerical value
32	9	≥ 0.0172; $\approx 1/32$ (size-9 set)	sizes exact; bound float-certified; value numerical

Open: closed forms for the non-stabiliser optima; the strong unextendibility constant for all five-element sets in $d = 16$ and beyond. *Scope*: all statements concern *Pauli-class* MUB sets; strongly unextendible sets of only three (non-Pauli) MUBs exist in many dimensions [5]. *Not claimed*: novelty of the weak-unextendibility census at $d = 16$, which may already be in Grassl's tables [3]. All certificates are in the repository (`data/*.json`) with the scripts that verify them.

References

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